

Gampa Sai Sasivardhan

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EDUCATION

- National Institute of Technology Warangal** 2023-2026
B.Tech in Electronic and Communication Engineering (Minor Degree) CGPA: 8.83
- National Institute of Technology Warangal** 2022-2026
B.Tech in Chemical Engineering (Major Degree) CGPA: 7.60
- Narayana College** 2020 - 2022
Board of Intermediate Education, Telengana Percentage: 95.2%
- Pallavi Model School** 2019 - 2020
Central Board of Secondary Education Percentage: 92.8%

EXPERIENCE

- Eternal Robotics** May 2025 - Jul 2025
Project Intern Hyderabad , India
 - Implemented an automated OCR and image classification pipeline using YOLO and PaddleOCR, improving text recognition accuracy by over 35% for industrial product labels, including both printed and low-contrast engraved text.
 - Built batch and GUI-based perspective correction tools in OpenCV (Tkinter GUI), enabling real-time image correction and enhancing OCR preprocessing speed by 0.22s per image, deployed for quality assurance automation.
- Mowito Robotics** Jun 2024 - Jul 2024
Project Intern Bengaluru , India
 - Developed an eccentricity checking machine, involving precision engineering and ROS2 integration with STM32 using OpenCV.
 - Designed a robotic conveyor system with substantially improved automation efficiency, utilizing STM32 Nucleo integration.
- Robotics Club NITW** Jan 2023 - Apr 2026
Secretary and Project Incharge Warangal , Telengana , India
 - Developed advanced robotic arms and AGVs with measurable improvements in operational efficiency, using ROS2 and precision engineering techniques.
 - Led the team in eYantra and Robocon, showcasing innovative robotic solutions, focusing on autonomous navigation and object manipulation.

PROJECTS

- Dynamic Live Location Tracking and Stop Request System for E-Carts** Nov 2024 - Dec 2024
Created a real-time GPS tracking system for efficient location updates.
 - Developed a real-time GPS tracking system with notably faster response times through Firebase-powered location updates.
 - Utilized ESP8266 and the TinyGPS++ library for seamless Wi-Fi communication and accurate GPS data parsing, achieving full data accuracy while storing and retrieving coordinates using Firebase Realtime Database.
 - Architected a system for instantaneous GPS coordinate sharing via Wi-Fi, substantially enhancing emergency response efficiency.
- Moon Rover** Jan 2024 - Aug 2024
Participated in ISRO's robotic competition called IROc-U.
 - Developed and executed an advanced autonomous rover solution with auto-navigation and robotic arm capabilities, achieving notable improvements in operational efficiency using Computer Vision and PID algorithms.
 - Developed a V-SLAM navigation model with meaningfully improved accuracy using Intel RealSense Depth Camera on Nvidia Jetson Nano for image processing.
 - Implemented PID algorithm for noticeably smoother robotic arm movement and speed control in sandy terrains using Raspberry Pi, achieving precise object picking with Inverse Kinematics.

•Waste Management Robot (WMR)

Dec 2023

Efficient waste management robot using Arduino IDE for sensor input and 3-axis robotic arm for segregation.

- Designed a sensor pad capable of reliably identifying waste types such as plastic, metal, paper, and wet waste using sensors and Arduino Mega.
- Developed a 3-axis robotic arm with substantially improved segregation efficiency for sorting waste into respective dustbins.
- Engineered an omnidirectional rover with a significant holding capacity using Arduino Uno, improving overall waste handling capability.

•5-Axis Robotic Arm

Jan 2024

Advanced 5-axis robotic arm employs OpenCV for color recognition and inverse kinematics for precise object placement

- Created a 5-axis robotic arm using high-torque servo motors, achieving high precision in segregation applications.
- Leveraged advanced OpenCV color recognition to achieve strong accuracy in object identification, employing inverse kinematics for precise placement.
- Executed PID algorithm for noticeably smoother servo movement, ensuring precise control with minimal error and enhancing overall performance.

•Autonomous Mobile Robot (AMR)

Oct 2024

Developed an autonomous mobile robot (AMR) using ROS2 for navigation, obstacle avoidance, and path planning.

- Implemented SLAM (Gmapping) for real-time mapping and localization, improving the robot's navigation accuracy.
- Integrated LiDAR, ultrasonic sensors, and cameras for obstacle detection and avoidance, enhancing real-time decision-making.
- Utilized OpenCV for camera-based object detection, optimizing path planning and navigation in dynamic environments.

TECHNICAL SKILLS AND INTERESTS

Languages: Python , Javascript , C++ , Embedded C, R, HTML , CSS

Developer Tools: VScode, AutoCAD, Fusion360, Jupyter Notebook, JetBrains, Arduino IDE, Microchip Studio, RPi OS, Blynk IoT, RemoteXY , Android Studio.

Databases: SQL, MongoDB

Technical Expertise : Robotics, Electronics, Internet of Things(IoT), Embedded Systems , Sensor Integration , ROS2 (Robot Operating System 2), Machine Learning, Mechanical Design , Web Development , App Development

Additional Projects: Quad copter , CNC Machine , Burglary Alarm System , Remote Controlled Car

Soft Skills: Effective communication , Leadership , Time management , Teamwork , Creativity, Problem Solving , Adaptability

Coursework: Electronic Engineering, IC Applications, Communication Systems, Internet of Things(IoT), Microprocessors and micro-controllers, Digital Electronics, AutoCAD, Process Instrumentation, Problem Solving and computer programming, MATLAB

Areas of Interest: Industrial and Mobile Robotics

POSITIONS OF RESPONSIBILITY

•Technical Event Co-ordinator, Technozion

Oct 2025

•Chairperson, IEEE SB Robotics and Automation Society(RAS) NITW

Aug 2025 - Dec 2025

•Genral Secretary, Robotics Club NITW

Apr 2025 - Apr 2026

ACHIEVEMENTS

•First Prize: ECE's Hackathon NITW

November 2023

•4th Prize: MJ College Hackathon

December 2023

•Top 150 Teams in ISRO IROC Rover Making Competition

January 2024

•Best Innovator and Best Innovation: Innovation Garage Ideathon

January 2025

•First Prize: Hitachi's Innothon 2.0

February 2025

•Special Prize: Hitachi's Innothon 3.0

March 2026

•Patent Recieved: Designed and Developed an EV GPS Locator. (Patent No: 587491)

April 2026